Beyond Topology: A Morphological Symmetry Graph Representation for Locomotion Policy Learning

Sizhe Wei[1*], **Xulin Chen**[2*], **Fengze Xie**[3], **Garrett Ethan Katz**[2], **Zhenyu Gan**[2], **Lu Gan**[1†]

[1]Georgia Institute of Technology [2]Syracuse University [3]California Institute of Technology

{swei,lgan}@gatech.edu {xchen168,gkatz01,zgan02}@syr.edu fxxie@caltech.edu

lunarlab-gatech.github.io/MS-PPO/

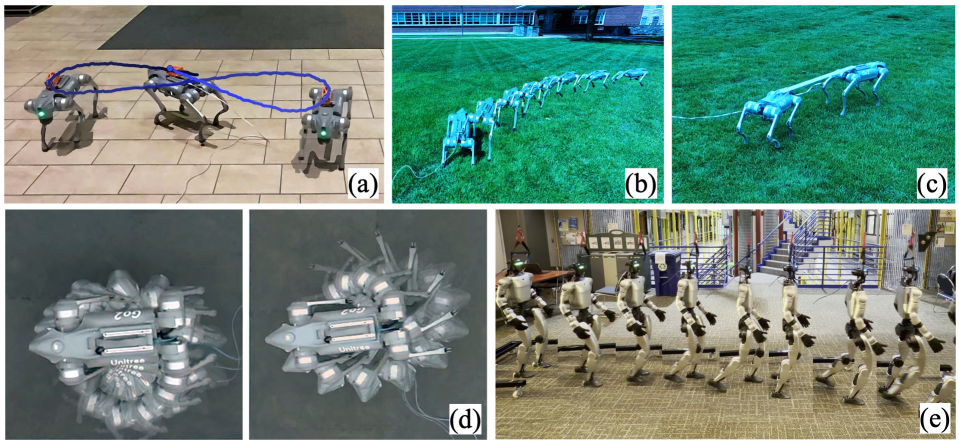


Figure 1: Go2 and G1 deployment: **(a)** Figure-8 tracking (Sec. 4.1); **(b)–(c)** locomotion and collaboration under rear-right calf failure (Sec. 4.2); **(d)** generalization to unseen mirrored commands (Sec. 4.1); and **(e)** zero-shot G1 locomotion (Sec. 4.3).

Abstract:

Reinforcement learning has enabled impressive locomotion on articulated robots, but generic policy networks leave robot structure implicit, while topology-based graphs do not specify how physical quantities transform across symmetric body parts. We introduce a morphological symmetry graph representation that jointly encodes kinematic connectivity and morphology-induced permutations and sign changes, yielding a graph actor with an equivariant action mean and an invariant critic. We instantiate it as `MS-PPO` with PPO on Unitree Go2 and `MS-SAC` with FastSAC on Unitree G1. Evaluation covers Go2 command tracking, mirrored-command generalization, and asymmetric joint failures, together with G1 29-DoF locomotion and zero-shot sim-to-real deployment. Our approach achieves the lowest mean tracking errors in simulation and hardware among the evaluated baselines, and the highest mean Go2 returns across three joint-failure types, using 34.6% of PPO's parameters. These results highlight morphological structure as a useful prior for real-world RL, supporting accurate tracking and robust zero-shot deployment from simulation-trained policies.

Keywords: Morphological Symmetry, Graph Representation, Robot Locomotion

1 Introduction

Reinforcement learning (RL) [1] enables capable control of humanoids [2, 3], quadrupeds [4], and dexterous hands [5], supported by high-fidelity simulation and parallel training [6]. For real-world RL, simulation offers a route to deployment with fewer on-robot learning interactions. Yet

[*] Equal contribution. [†] Corresponding author.

generic networks flatten observations and leave known physical structure to be learned implicitly. This motivates using robot morphology as an explicit architectural prior. We ask whether known morphological structure can support robust sim-to-real locomotion without hardware fine-tuning.

A topology-informed graph represents joints and links, supporting information flow along kinematic connections and parameter sharing across repeated limbs [7, 8, 9, 10, 11]. Connectivity alone, however, does not specify how observations and actions transform between symmetric joints. Topological sharing alone does not ensure physically correct mirrored actions.

Symmetry-aware RL uses mirrored data augmentation [12], reward shaping [13], or equivariant architectures [14, 15, 16]. The first two encourage symmetry through data or objectives; the third enforces coordinate transformations directly. These constraints alone do not specify information flow along the kinematic structure, motivating a representation that combines both priors.

We introduce a morphological symmetry graph representation with signed sagittal-reflection actions on observations and actions [17, 18], an equivariant action mean, and an invariant critic. These constraints are enforced directly by the network architecture. Instantiations use PPO on Go2 (`MS-PPO`) and FastSAC on G1 (`MS-SAC`), with separately trained policies. Experiments cover command tracking, mirrored-command generalization, Go2 single-joint failure compensation, and G1 29-DoF locomotion. Our method achieves the lowest mean tracking errors in simulation and hardware and the highest mean Go2 returns across three fault types, with 34.6% of PPO's parameters. By incorporating known robot structure before deployment, our approach supports robust zero-shot transfer and offers a promising route toward reduced reliance on real-world adaptation.

2 Related Work

2.1 Topology-Informed Robot Policy

Topology-informed policies encode kinematic structure to improve sample efficiency [9], adaptability [19], and cross-embodiment generalization [10]. Graph policies pass messages along joints and links and share parameters across repeated body parts, supporting modular control [20, 8]. Transformer attention can likewise be viewed as graph message passing [21, 22]. Body Transformer [9] masks attention by topology; MetaMorph [10] encodes URDF-derived morphology in positional embeddings. EAT [19] studies locomotion under changing morphology, while related methods use joint tokens and structural attention biases [23, 11]. MI-HGNN [24] uses topology-informed graph representations for contact perception. We adapt this structural prior to policy learning and incorporate morphology-consistent coordinate transformations [18].

2.2 Symmetric Legged Locomotion

Symmetry promotes regular locomotion and can improve learning efficiency [25]. Prior approaches use symmetric hybrid dynamics [26, 27], mirrored data augmentation [12], reward shaping [13], or symmetry-preserving architectures [14, 15]. Augmentation and reward shaping encourage approximate equivariance, whereas equivariant architectures enforce specified state and action transformations. Our representation integrates these transformations with topology-aware information flow, making both priors explicit in the policy graph.

3 Methodology

Our representation combines kinematic connectivity with symmetry-induced coordinate transformations (Figure 2). An encoder maps observations to graph features, an equivariant backbone propagates information along the topology, and readouts produce an equivariant action mean or invariant value estimate. We instantiate this construction with PPO on Go2 and FastSAC on G1.

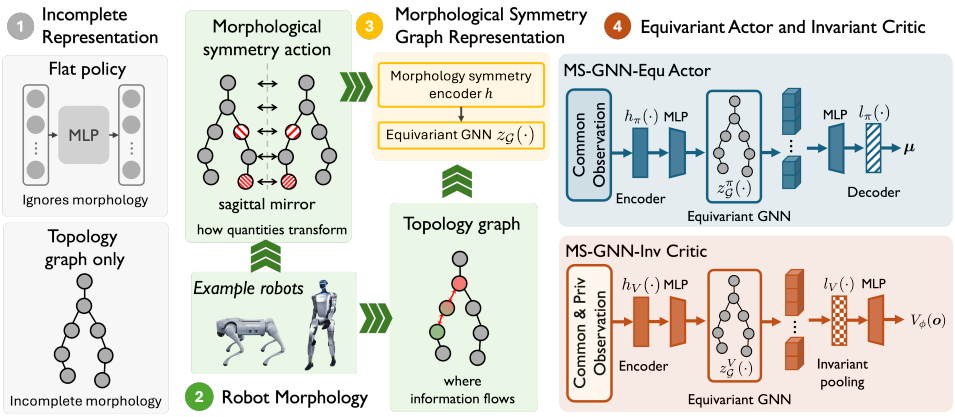


Figure 2: Overview of the morphological symmetry graph representation. Kinematic connectivity specifies where information flows, while morphology-induced permutations and sign changes specify how signals transform. The MS-GNN-Equ actor maps reflected observations to reflected action means; the MS-GNN-Inv critic produces the same value for reflected inputs.

3.1 Symmetry Actions and MDP Assumptions

We consider sagittal reflection $G = \mathbb{C}_2 = \{e, m\}$. On each vector space $\mathcal{Y}$, the representation $\rho_{\mathcal{Y}} : G \to \mathrm{GL}(\mathcal{Y})$ satisfies $\rho_{\mathcal{Y}}(gh) = \rho_{\mathcal{Y}}(g)\rho_{\mathcal{Y}}(h)$. In particular, $\rho_{\mathcal{O}}$ and $\rho_{\mathcal{A}}$ transform observations and actions by morphology-induced permutations and sign changes. We denote the transformation of the full MDP state by $T_g : \mathcal{S} \to \mathcal{S}$.

The augmented MDP state includes physical state, commands, persistent faults, and randomized environment parameters; T_g acts jointly on their symmetry-dependent components. For every $g \in G$ and measurable $B \subseteq \mathcal{S}$, we assume

$$\begin{aligned} p_0(T_g B) &= p_0(B), \\ r(T_g s, \rho_{\mathcal{A}}(g)a) &= r(s, a), \\ P(T_g B \mid T_g s, \rho_{\mathcal{A}}(g)a) &= P(B \mid s, a), \end{aligned} \tag{1}$$

where p_0, r, and P are the initial-state distribution, reward, and transition kernel. Disturbances, randomization, observations, and termination must respect reflection. These are distributional assumptions: uniformly sampled mirrored faults can preserve symmetry despite asymmetric individual failures. They motivate invariant values for optimal and distributionally equivariant policies.

3.2 Robot-to-Graph Encoding

We construct a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ from the kinematic tree, with nodes $\mathcal{V}$ representing body components or joint groups, and edges $\mathcal{E}$ representing the connectivity. Actor inputs $o \in \mathcal{O}$ comprise the available proprioceptive observations and commands, with temporal histories where used. Group actions apply framewise to histories; finite history is not assumed to make the latent state fully observable.

Following [18], the encoder absorbs coordinate sign changes so that graph features transform by reindexing. Paired body parts use sign masks; base features $b \in \mathcal{B}$ are lifted to two virtual nodes $(b, \rho_{\mathcal{B}}(m)b)$. Reflection swaps this pair because $\rho_{\mathcal{B}}(m)^2 = I$, preserving reflection-odd information at the otherwise fixed base. Polar vectors (linear velocity, gravity) and axial vectors (angular velocity) require distinct masks; joint signs follow the mirrored joint frames.

The actor encoder h_π and decoder l_π satisfy

$$\begin{aligned} h_\pi(\rho_{\mathcal{O}}(g)o) &= P_g h_\pi(o), \\ l_\pi(P_g Z) &= \rho_{\mathcal{A}}(g) l_\pi(Z). \end{aligned} \tag{2}$$

Node-type projections and L sum-aggregation graph layers share parameters consistently across symmetry orbits, yielding an equivariant graph neural network $z_{\mathcal{G}}$ on a feature space X

$$z_{\mathcal{G}}(P_g X) = P_g z_{\mathcal{G}}(X), \tag{3}$$

where the node permutation P_g preserves connectivity and node/edge types on the graph $\mathcal{G}$. Changing the embodiment changes the graph, feature assignments, and coordinate transformations while preserving this construction.

3.3 Equivariant Actor and Invariant Critic

We use separately parameterized graph backbones $z^{\pi}_{\mathcal{G}}$ and $z^{V}_{\mathcal{G}}$ for the actor and critic, respectively. The action mean is the composition $\mu_\theta = l_\pi \circ z^{\pi}_{\mathcal{G}} \circ h_\pi$. Equations (2)–(3) give

$$\mu_\theta(\rho_{\mathcal{O}}(g)o) = \rho_{\mathcal{A}}(g)\mu_\theta(o). \tag{4}$$

Thus, deterministic deployment maps reflected observations to reflected actions by construction.

For the PPO critic, $u = [o, o_{\text{priv}}] \in \mathcal{U}$ includes privileged observations with action $\rho_{\mathcal{U}}$. A separate equivariant backbone defines $Z = z^{V}_{\mathcal{G}}(h_V(u))$ and $V_\phi(u) = l_V(Z)$. For each mirrored node pair $(i, m(i))$, the invariant readout concatenates $Z_i + Z_{m(i)}$, $|Z_i - Z_{m(i)}|$, and $(Z_i - Z_{m(i)})^2$ in fixed orbit order, followed by an MLP. Swapping a pair leaves these statistics unchanged, giving

$$V_\phi(\rho_{\mathcal{U}}(g)u) = V_\phi(u). \tag{5}$$

This constrains the critic, not every exploratory policy's value.

3.4 PPO and FastSAC Instantiations

For Go2, `MS-PPO` uses PPO [28] with

$$\pi_\theta(\cdot \mid o) = \mathcal{N}\big(\mu_\theta(o), \operatorname{diag}(\sigma^2)\big)\,. \tag{6}$$

The standard deviations are learned independently. Consequently, the action mean is exactly equivariant, but the training policy distribution is not constrained to be equivariant: this would additionally require $\Sigma(\rho_{\mathcal{O}}(g)o) = \rho_{\mathcal{A}}(g)\Sigma(o)\rho_{\mathcal{A}}(g)^\top$.

For G1, we use FastSAC [29] to accelerate training, retaining the equivariant actor design. The critic estimates an action value and is constrained under joint observation-action transformation,

$$Q_\phi(\rho_{\mathcal{U}}(g)u, \rho_{\mathcal{A}}(g)a) = Q_\phi(u, a). \tag{7}$$

4 Experiments

We evaluate command tracking, single-joint fault compensation, and sim-to-real deployment on Unitree Go2 [30] and G1 [31]. Policies are trained in simulation on NVIDIA RTX 5090 and L40S GPUs. Hardware tests evaluate tracking, generalization, and fault compensation after simulation training.

Baselines compare topology and symmetry priors. PPO uses MLP actor-critic networks [28]; PPO-Aug adds mirrored data [12]. PPO-EMLP uses a $\mathbb{C}_2$-equivariant action mean and invariant critic [14, 17, 32]. MI-PPO adapts the morphology-informed graph representation of MI-HGNN [24] to policy learning [7], with unsigned leg permutations; MI-PPO-Aug adds signed $\mathbb{C}_2$ augmentation without changing that constraint. The four-layer Body Transformer (BoT) [9] incorporates topology through body-structured attention.

Evaluation protocol. All seven Go2 methods use seeds 1, 4, 23, 42, and 123. Eligibility rate (ER) is the fraction of final checkpoints retaining first-episode tracking samples after a 50-step no-fault Figure-8 warm-up. A seed is excluded only if all 16 environments terminate before contributing post-warm-up samples, due to abnormal base contact. Figure-8 and fault metrics use the same eligible checkpoints; ER separately reports this selection rate.

Metrics. Linear tracking error (Lin. TE) averages planar velocity-error norms; angular tracking error (Ang. TE) averages absolute yaw-rate errors. Go2 reward is undiscounted first-episode return, including warm-up and terminal reward. G1 reports mean per-step reward.

Table 1: Go2 simulation results. Topo. indicates an explicit topology prior; Symm. identifies the encoded or augmented symmetry. Task metrics use eligible seeds: †two for PPO-EMLP, ‡four for MI-PPO-Aug, and five for each other method. Entries report mean ± population standard deviation across eligible seeds for Figure-8 and across eligible seed-leg pairs for each joint-failure type. Figure-8 tracking errors exclude the 50-step warm-up; rewards include it.

Method	ER ↑	Topo.	Symm.	*walk-figure-8*			*joint zero-torque*			Num. Params. ↓
				Lin. TE ↓ (m/s)	Ang. TE ↓ (rad/s)	Reward ↑	Hip reward ↑	Thigh reward ↑	Calf reward ↑	
PPO [28]	5/5			0.084 ± 0.015	0.150 ± 0.007	13.295 ± 1.946	4.797 ± 2.158	10.244 ± 1.634	2.217 ± 1.884	2, 113, 945
PPO-Aug [12]	5/5		$\mathbf{C}_2$-Aug	0.077 ± 0.020	0.146 ± 0.014	14.464 ± 0.483	7.913 ± 2.150	11.869 ± 1.741	2.907 ± 2.428	2, 113, 945
PPO-EMLP [14]†	2/5		$\mathbf{C}_2$	0.086 ± 0.006	0.130 ± 0.008	14.206 ± 0.179	4.599 ± 3.134	10.763 ± 0.703	2.332 ± 2.024	1, 056, 979
BoT [9]	5/5	✓		0.170 ± 0.029	0.200 ± 0.030	11.910 ± 0.670	0.738 ± 1.040	1.827 ± 2.071	1.048 ± 1.030	2, 041, 765
MI-PPO [24, 7]	5/5	✓	$\mathbf{S}_4$	0.089 ± 0.014	0.165 ± 0.007	14.442 ± 0.341	5.929 ± 2.821	10.802 ± 2.087	2.291 ± 1.984	864, 142
MI-PPO-Aug‡	4/5	✓	$\mathbf{S}_4$ ($\mathbf{C}_2$-Aug)	0.293 ± 0.170	0.603 ± 0.266	6.945 ± 5.135	2.578 ± 4.311	3.597 ± 5.835	1.627 ± 2.421	864, 142
`MS-PPO` (Ours)	5/5	✓	$\mathbf{C}_2$	$\mathbf{0.061 \pm 0.014}$	$\mathbf{0.101 \pm 0.011}$	$\mathbf{14.679 \pm 1.736}$	$\mathbf{9.418 \pm 1.606}$	$\mathbf{12.719 \pm 1.456}$	$\mathbf{4.023 \pm 3.045}$	**732, 046**

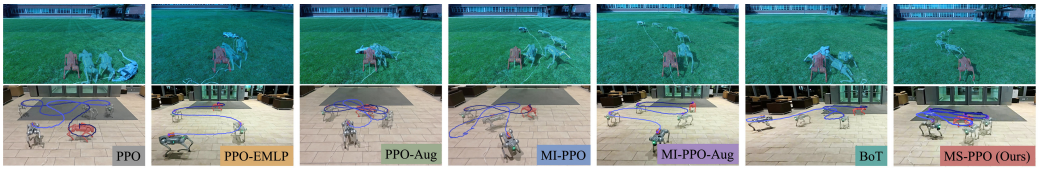


Figure 3: Visualized hardware results. The top row shows *joint zero-torque*, where the robot is commanded to track $v_x^{\text{cmd}} = 1$m/s with the rear-right calf joint disabled. The bottom row shows *walk-figure-8*, with each method evaluated over three consecutive figure-8 cycles. Blue curves show the robot trajectories, and red indicates the initial positions. MS-PPO maintains a more stable path in *joint zero-torque* and achieves better symmetric command tracking in *walk-figure-8* compared to baselines. Quantitative errors are reported in Table 2.

4.1 Quadrupedal Figure-8 Walking

The Go2 graph assigns gravity and commands to the base, and joint states, previous actions, and phases to joint nodes. Critic inputs add reflection-invariant friction and restitution.

Go2 training uses *Walk-These-Ways* [33] with zero-torque failures [34]. Each reset samples nominal operation with probability 0.7 or a uniformly chosen single-joint failure with probability 0.3; the selected joint's commanded torque remains zero throughout the episode. All methods use this mixture and receive 30 frames of proprioception and previous actions (0.6 s at 50 Hz), without fault identity. History provides evidence of the latent actuator condition, but does not guarantee its identification.

The *walk-figure-8* task alternates left and right turns using velocity commands derived from $x(t) = 2\sin(2\pi t/20)$ and $y(t) = \sin(4\pi t/20)$, in meters and seconds. The reference has a 20-s period and $4\,\text{m} \times 2\,\text{m}$ envelope; the realized path depends on tracking and survival. Simulation covers one period; hardware covers three consecutive periods (60 s) per method. `MS-PPO` achieves the lowest mean tracking errors in both settings and the highest mean simulation return (Tables 1 and 2; Figure 3).

Relative to the strongest baseline for each metric, `MS-PPO` reduces linear/angular errors by 20.8%/22.3% in simulation and 2.8%/9.5% on hardware. These gains support coordinated tracking across mirrored turns and show that the representation remains useful after transfer.

Generalization to mirrored commands. In a separate experiment, Go2 trotting and pronking policies train with $c_x \in [-1, 1]$ m/s, $c_y \in [0, 0.6]$ m/s, and $c_\omega \in [0, 1]$ rad/s. We test unseen mirrored commands by reversing c_y and c_ω. Both gaits retain comparable simulation tracking errors. On hardware, the trotting policy sustains 15-s in-place turns at $c_x = c_y = 0$, $c_\omega = \pm 1$ rad/s without retraining (see video). This demonstrates generalization to reflection-related commands.

4.2 Joint Failure Tolerance

The *joint zero-torque* experiment commands forward walking at 1 m/s under each of 12 unobserved single-joint failures, with 100 environments per seed and fault for up to 1000 steps. Figure 4 reports

Table 2: Mean ± standard deviation of Go2 hardware tracking errors. Figure-8 aggregates three consecutive cycles; zero-torque results use the longer-surviving of two trials per method.

Method	*walk-figure-8*		*joint zero-torque*	
	Lin. TE ↓ (m/s)	Ang. TE ↓ (rad/s)	Lin. TE ↓ (m/s)	Ang. TE ↓ (rad/s)
PPO	0.222 ± 0.116	0.332 ± 0.240	1.203 ± 0.247	0.486 ± 0.378
PPO-Aug	0.217 ± 0.102	0.266 ± 0.146	1.209 ± 0.183	0.892 ± 0.927
PPO-EMLP	0.301 ± 0.143	0.291 ± 0.182	1.673 ± 1.810	0.714 ± 0.553
BoT	0.273 ± 0.141	0.333 ± 0.225	1.241 ± 0.222	0.900 ± 0.473
MI-PPO	0.239 ± 0.122	0.308 ± 0.177	0.825 ± 0.287	0.580 ± 0.550
MI-PPO-Aug	0.256 ± 0.115	0.243 ± 0.167	0.543 ± 0.337	0.538 ± 0.473
MS-PPO (Ours)	$\mathbf{0.211 \pm 0.112}$	$\mathbf{0.220 \pm 0.125}$	$\mathbf{0.405 \pm 0.263}$	$\mathbf{0.246 \pm 0.216}$

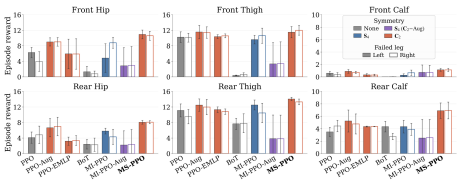


Figure 4: Go2 joint-failure tolerance in simulation. Each group reports first-episode return for one disabled joint. Bars show means and standard deviations across eligible seeds.

individual faults; Table 1 aggregates first-episode returns across legs. MS-PPO achieves the highest mean hip-, thigh-, and calf-failure returns.

Symmetry and local compensation. Reflection maps each unilateral fault to its opposite-side counterpart. Uniform fault sampling can preserve distributional symmetry while allowing side-specific compensation within a rollout. The 30-frame history supplies fault evidence, and graph connectivity organizes information flow between mechanically connected joints.

Training success and representation. All five MS-PPO checkpoints pass the eligibility screen, versus two for PPO-EMLP and four for MI-PPO-Aug. Both PPO-EMLP and MS-PPO constrain the mean by $\mathbb{C}_2$, but MS-PPO adds kinematic connectivity and structured sharing. Higher ER and returns with fewer parameters (732,046 versus 1,056,979) support these priors empirically.

MI-PPO's unsigned $\mathbf{S}_4$ permutations omit spatial roles and coordinate signs required by physical reflection. MI-PPO-Aug doubles training samples with signed reflections but retains this architectural mismatch, and exhibits lower ER and fault returns. In contrast, MS-PPO combines signed reflection with kinematic connectivity to support position-dependent compensation.

Hardware fault tolerance. Go2 tests disable the rear-right calf (Figure 3, Table 2). To limit damage, we run two trials per method and report the longer-surviving trial for both visualization and metrics. In these selected trials, MS-PPO maintains sustained locomotion and the lowest mean errors, improving linear/angular tracking by 25.4%/49.4% over the respective strongest baselines. These are selected-trial results, not repeated-trial reliability estimates.

In the dual-robot demonstration (Figure 5), a rear robot drags or perturbs the faulted lead robot. MS-PPO maintains stable locomotion in the recorded sequence, providing qualitative evidence under combined failure and interaction disturbances.

4.3 Humanoid Whole-Body Locomotion

G1 experiments in *Holosoma* [35, 36] extend the representation to 29 actions: 12 leg, 3 waist, and 14 arm joints. We train G1 policies separately, without transferring Go2 weights.

Training framework. All seven G1 methods use FastSAC [29] on the same task. We retain the equivariant actor and adapt the invariant critic to SAC, yielding MS-SAC.

Whole-body graph. The 100-dimensional observation contains joint states, previous actions, base angular velocity, gravity, commands, and gait phases. A 14-node graph has two virtual bases, three nodes per leg (hip, knee, ankle), and three per arm (shoulder, elbow, wrist). The bases hold symmetry-related waist features, not duplicate physical bodies. Mirrored limbs share signed encoders and decoders; waist outputs respect each action's reflection parity.

Simulation and hardware evaluation. Simulation uses eight command groups, 80 environments, and 1000 steps. We aggregate groups within each seed and report mean ± standard deviation over five seeds. All deterministic actors deploy without fine-tuning; hardware tests cover 0.7 m/s forward

Table 3: G1 29-DoF locomotion. Simulation: mean ± std over five seeds and eight command groups. Hardware: mean ± std over three trials at 0.7 m/s forward, without fine-tuning.

Method	Simulation			Hardware	
	Reward ↑	Lin. TE ↓ (m/s)	Ang. TE ↓ (rad/s)	Lin. TE ↓ (m/s)	Ang. TE ↓ (rad/s)
SAC [29]	0.154 ± 0.002	0.395 ± 0.022	0.139 ± 0.009	0.185 ± 0.019	0.158 ± 0.011
SAC-Aug [12]	0.156 ± 0.001	0.374 ± 0.022	0.138 ± 0.009	0.191 ± 0.005	0.188 ± 0.017
SAC-EMLP [14]	0.153 ± 0.003	0.403 ± 0.041	0.136 ± 0.016	0.170 ± 0.004	0.140 ± 0.024
SAC-BoT [9]	0.147 ± 0.010	0.475 ± 0.117	0.254 ± 0.158	0.289 ± 0.015	0.154 ± 0.034
MI-SAC [24, 7]	0.153 ± 0.002	0.414 ± 0.046	0.147 ± 0.016	0.215 ± 0.017	0.258 ± 0.061
MI-SAC-Aug	0.152 ± 0.003	0.410 ± 0.045	0.156 ± 0.019	0.188 ± 0.017	0.222 ± 0.033
`MS-SAC` (ours)	$\mathbf{0.159 \pm 0.004}$	$\mathbf{0.338 \pm 0.037}$	$\mathbf{0.123 \pm 0.023}$	$\mathbf{0.146 \pm 0.009}$	$\mathbf{0.139 \pm 0.007}$

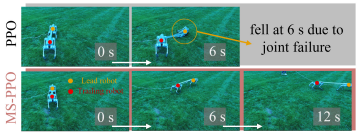


Figure 5: Dual-robot collaboration under joint failure. The lead robot tracks 1 m/s command with its rear-right calf joint disabled.

Table 4: Go2 training and inference computational cost. Training time is projected to 5k iterations on an L40S; batch-one inference is measured on an RTX 4090.

Method	Training (h)	Latency (ms)	FLOPs (10^6)	Memory (MiB)
PPO	**3.005**	$\mathbf{0.0348 \pm 0.0012}$	**2.113**	9.075
PPO-Aug	3.363	0.0350 ± 0.0023	**2.113**	9.075
PPO-EMLP	3.484	0.0467 ± 0.0026	**2.113**	41.864
BoT	10.814	0.6087 ± 0.0209	10.854	5.886
MI-PPO	7.751	0.2915 ± 0.0098	7.685	2.306
MI-PPO-Aug	12.101	0.2899 ± 0.0100	7.685	2.306
`MS-PPO`	6.010	0.2002 ± 0.0065	4.436	**1.823**

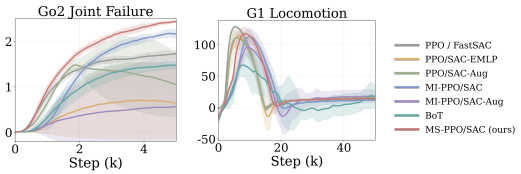


Figure 6: Training reward curves for Go2 locomotion with joint-failure exposure and G1 humanoid locomotion. G1 curves illustrate training dynamics under an adaptive penalty curriculum; increasing penalty weights contribute to the temporary reward decrease.

walking, with three trials per method. Planar velocity is reconstructed offline from foot kinematics and IMU measurements under a stance-foot assumption; the IMU provides yaw rate.

`MS-SAC` achieves the highest mean simulation reward and lowest mean tracking errors in simulation and hardware (Table 3). Relative to SAC-EMLP, hardware linear error decreases from 0.170 to 0.146 m/s (approximately 14%); angular means differ only slightly (0.140 versus 0.139 rad/s). With all methods deployed without hardware fine-tuning, these results demonstrate improved linear tracking from morphological priors under the tested command.

4.4 Training Efficiency Across Tasks

Figure 6 shows Go2 PPO and G1 FastSAC training curves. On Go2, `MS-PPO` reaches higher rewards earlier and retains the highest late-training reward. G1 uses an adaptive penalty curriculum: weights increase with episode length, contributing to temporary reward decreases. Methods can therefore face different penalties at the same step; these curves illustrate training dynamics rather than rank learning speed. G1 performance is assessed under the common evaluation in Table 3.

Table 4 reports Go2 computational cost. Projected training time is 6.010 h for `MS-PPO`, versus 3.005 h for PPO and 10.814 h for BoT. Fixed-topology matrix operations avoid dynamic scatter, whereas the evaluated BoT uses dense masked attention. At batch one on an RTX 4090, `MS-PPO` requires 4.436M FLOPs, 0.2002 ms, and 1.823 MiB peak memory. The latency is about 1% of a 20-ms control period on this GPU, not an end-to-end onboard measurement. Thus, `MS-PPO` trades higher training cost than MLPs for better Go2 learning progress and lower graph/Transformer inference cost.

4.5 Ablation

We examine the contributions of the equivariant MS-GNN actor and invariant MS-GNN critic by independently replacing each with an MLP (Table 5). Each configuration is evaluated over all 12 zero-torque joint-failure conditions using three training seeds. Relative

Table 5: Actor-critic architecture ablation under zero-torque joint failures. Each seed is first averaged over all 12 fault conditions, and results are reported as mean ± standard deviation across three random seeds.

Actor	Critic	Lin. TE ↓	Ang. TE ↓	Reward ↑
MLP	MLP	0.439 ± 0.023	0.567 ± 0.071	5.561 ± 0.317
MLP	Invariant MS-GNN	0.368 ± 0.029	0.463 ± 0.053	6.764 ± 0.559
Equivariant MS-GNN	MLP	0.304 ± 0.038	$\mathbf{0.333 \pm 0.037}$	8.146 ± 0.413
Equivariant MS-GNN	Invariant MS-GNN	$\mathbf{0.278 \pm 0.036}$	0.390 ± 0.080	$\mathbf{8.522 \pm 0.448}$

to the MLP actor-critic baseline,
replacing the MLP actor or critic alone increases mean reward from 5.561 to 8.146 (46.5%) or 6.764 (21.6%), respectively. Both replacements reduce tracking errors, with the actor yielding the larger performance gain. Combining the equivariant actor with the invariant critic achieves the highest mean reward and lowest mean linear tracking error. Compared with the equivariant actor with an MLP critic, the invariant critic raises reward to 8.522 (+4.6%) and reduces linear error to 0.278 (-8.6%), while angular error increases to 0.390 (+17.1%). Thus, the invariant critic provides an additional improvement in aggregate return and linear tracking, with a trade-off in angular tracking. Overall, the results support the combined actor-critic design, with the equivariant actor providing the larger gain and the invariant critic further improving the aggregate objective.

5 Conclusion

We introduced a morphological symmetry graph representation that jointly encodes kinematic connectivity and physical coordinate transformations. Instantiated as `MS-PPO` on Go2 and `MS-SAC` on G1, it supports command tracking, mirrored-command generalization, single-joint fault compensation, and zero-shot humanoid deployment. By building known robot structure into policy learning, our approach translates simulation training into capable real-world behavior, motivating morphological priors as a route toward reduced reliance on post-deployment adaptation.

6 Limitations

Validation covers manually specified bilateral $\mathbf{C}_2$ symmetry on Go2 and G1, with separately trained policies rather than cross-embodiment policy transfer. G1 hardware tests cover one forward command; faults cover single-joint zero-torque failures. Richer or approximate symmetries, dynamic whole-body skills, and broader hardware conditions remain untested. Our real-world evaluation focuses on zero-shot deployment. Future work will quantify on-robot adaptation requirements and automate graph construction.

References

[1] R. S. Sutton and A. G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2 edition, 2018.

[2] Y. Ze, Z. Chen, J. P. Araújo, Z. ang Cao, X. B. Peng, J. Wu, and C. K. Liu. Twist: Teleoperated whole-body imitation system. *arXiv preprint arXiv:2505.02833*, 2025.

[3] I. Radosavovic, T. Xiao, B. Zhang, T. Darrell, J. Malik, and K. Sreenath. Real-world humanoid locomotion with reinforcement learning. *Science Robotics*, 9(89):eadi9579, 2024.

[4] T. Miki, J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter. Learning robust perceptive locomotion for quadrupedal robots in the wild. *Science Robotics*, 7(62):eabk2822, 2022.

[5] H. Qi, A. Kumar, R. Calandra, Y. Ma, and J. Malik. In-hand object rotation via rapid motor adaptation. In *Proceedings of the Conference on Robot Learning*, pages 1722–1732, 2023.

[6] N. Rudin, D. Hoeller, P. Reist, and M. Hutter. Learning to walk in minutes using massively parallel deep reinforcement learning. In *Proceedings of the Conference on Robot Learning*, pages 91–100, 2022.

[7] T. Wang, R. Liao, J. Ba, and S. Fidler. Nervenet: Learning structured policy with graph neural networks. In *Proceedings of the International Conference on Learning Representations*, 2018.

[8] J. Whitman, M. Travers, and H. Choset. Learning modular robot control policies. *IEEE Transactions on Robotics*, 39(5):4095–4113, 2023.

[9] C. Sferrazza, D.-M. Huang, F. Liu, J. Lee, and P. Abbeel. Body transformer: Leveraging robot embodiment for policy learning. In *Proceedings of the Conference on Robot Learning*, pages 3407–3424, 2025.

[10] A. Gupta, L. Fan, S. Ganguli, and L. Fei-Fei. Metamorph: Learning universal controllers with transformers. In *Proceedings of the International Conference on Learning Representations*, 2022.

[11] A. Patel and S. Song. Get-zero: Graph embodiment transformer for zero-shot embodiment generalization. In *Proceedings of the IEEE International Conference on Robotics and Automation*, pages 14262–14269, 2025.

[12] M. Mittal, N. Rudin, V. Klemm, A. Allshire, and M. Hutter. Symmetry considerations for learning task symmetric robot policies. In *Proceedings of the IEEE International Conference on Robotics and Automation*, pages 7433–7439, 2024.

[13] J. Ding, X. Chen, G. E. Katz, and Z. Gan. Towards dynamic quadrupedal gaits: A symmetry-guided rl hierarchy enables free gait transitions at varying speeds. *arXiv preprint arXiv:2403.10723*, 2024.

[14] Z. Su, X. Huang, D. F. O. Apraez, Y. Li, Z. Li, Q. Liao, G. Turrisi, M. Pontil, C. Semini, Y. Wu, and K. Sreenath. Leveraging symmetry in rl-based legged locomotion control. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 6899–6906, 2024.

[15] B. Nie, Y. Zhang, R. Jin, Z. Cao, H. Lin, X. Yang, and Y. Gao. Coordinated humanoid robot locomotion with symmetry equivariant reinforcement learning policy. *arXiv preprint arXiv:2508.01247*, 2025.

[16] Z. Li, Y. Jin, D. O. Apraez, C. Semini, P. Liu, and G. Chalvatzaki. Morphologically symmetric reinforcement learning for ambidextrous bimanual manipulation. *arXiv preprint arXiv:2505.05287*, 2025.

[17] D. O. Apraez, G. Turrisi, V. Kostic, M. Martin, A. Agudo, F. Moreno-Noguer, M. Pontil, C. Semini, and C. Mastalli. Morphological symmetries in robotics. *International Journal of Robotics Research*, 44(10-11):1743–1766, 2025.

[18] F. Xie, S. Wei, Y. Song, Y. Yue, and L. Gan. Morphological-symmetry-equivariant heterogeneous graph neural network for robotic dynamics learning. In *Proceedings of the Learning for Dynamics & Control Conference*, pages 1392–1405, 2025.

[19] C. Yu, W. Zhang, H. Lai, Z. Tian, L. Kneip, and J. Wang. Multi-embodiment legged robot control as a sequence modeling problem. In *Proceedings of the IEEE International Conference on Robotics and Automation*, pages 7250–7257, 2023.

[20] W. Huang, I. Mordatch, and D. Pathak. One policy to control them all: Shared modular policies for agent-agnostic control. In *Proceedings of the International Conference on Machine Learning*, pages 4455–4464, 2020.

[21] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. Attention is all you need. In *Proceedings of the Advances in Neural Information Processing Systems Conference*, pages 5998–6008, 2017.

[22] M. M. Bronstein, J. Bruna, T. Cohen, and P. Veličković. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges, 2021. URL https://arxiv.org/abs/2104.13478.

[23] S. Hong, D. Yoon, and K.-E. Kim. Structure-aware transformer policy for inhomogeneous multi-task reinforcement learning. In *Proceedings of the International Conference on Learning Representations*, 2022.

[24] D. Butterfield, S. S. Garimella, N.-J. Cheng, and L. Gan. Mi-hgnn: Morphology-informed heterogeneous graph neural network for legged robot contact perception. In *Proceedings of the IEEE International Conference on Robotics and Automation*, pages 10110–10116, 2025.

[25] F. Abdolhosseini, H. Y. Ling, Z. Xie, X. B. Peng, and M. Van de Panne. On learning symmetric locomotion. In *Proceedings of the 12th ACM SIGGRAPH Conference on Motion, Interaction and Games*, pages 1–10, 2019.

[26] R. Altendorfer, D. E. Koditschek, and P. Holmes. Stability analysis of legged locomotion models by symmetry-factored return maps. *International Journal of Robotics Research*, 23 (10-11):979–999, 2004.

[27] H. Razavi, A. M. Bloch, C. Chevallereau, and J. W. Grizzle. Symmetry in legged locomotion: a new method for designing stable periodic gaits. *Autonomous Robots*, 41(5):1119–1142, 2017.

[28] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

[29] Y. Seo, C. Sferrazza, J. Chen, G. Shi, R. Duan, and P. Abbeel. Learning sim-to-real humanoid locomotion in 15 minutes. *arXiv preprint arXiv:2512.01996*, 2025.

[30] Unitree Go2. https://www.unitree.com/go2, . Accessed: 2025-06-25.

[31] Unitree G1. https://www.unitree.com/g1, . Accessed: 2026-05-01.

[32] G. Cesa, L. Lang, and M. Weiler. A program to build e(n)-equivariant steerable cnns. In *Proceedings of the International Conference on Learning Representations*, 2022.

[33] G. B. Margolis and P. Agrawal. Walk these ways: Tuning robot control for generalization with multiplicity of behavior. In *Proceedings of the Conference on Robot Learning*, pages 22–31, 2023.

[34] K. Bao, C. Li, Y. As, A. Krause, and M. Hutter. Toward task generalization via memory augmentation in meta-reinforcement learning. *arXiv preprint arXiv:2502.01521*, 2025.

[35] Amazon FAR, P. Abbeel, J. Chen, R. Duan, A. Escontrela, M. Gandhi, S. Gundry, X. Huang, A. Kanazawa, T. Lewicki, J. Li, K. Liu, C. Rosenthal, Y. Seo, C. Sferrazza, G. Shi, L. Shih, J. Tseng, Z. Wu, L. Yang, B. Yi, and Y. Zhang. Holosoma.

[36] M. Mittal, P. Roth, J. Tigue, et al. Isaac Lab - A GPU-Accelerated Simulation Framework for Multi-Modal Robot Learning. *arXiv preprint arXiv:2511.04831*, 2025. doi:10.48550/arXiv.2511.04831. URL https://arxiv.org/abs/2511.04831.